

Before the Levy: Environmental Policy Announcements and Denmark's Beef Market

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Abstract

Environmental policies are often announced years before they take effect. This paper examines whether firms and consumers adjust during that interval, using Denmark's June 2024 announcement of a tax on livestock emissions. A model with costly adjustment links current production plans to expected future costs and predicts higher prices before tax collection. I examine this prediction using supermarket prices and official consumer price indices. Difference-in-differences estimates indicate beef-price increases of about 3 and 6 log points, respectively, although synthetic difference-in-differences estimates are less precise. Monthly cattle slaughter in Denmark and other EU countries finds a negative but imprecise output response that is sensitive to the pre-treatment window. The evidence is consistent with anticipatory adjustment but does not establish supply contraction as its mechanism. Concurrent changes in cattle regulation also limit attribution to the tax announcement alone. The findings illustrate both the potential reach of environmental-policy announcements and the identification challenges involved in measuring their effects.

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1 Introduction

Environmental policy can change economic decisions well before it changes firms' current tax liabilities. Producers expecting a future emissions charge may revise investment, procurement, and output plans when the policy is announced. Consumers may then face different prices during the years that separate agreement from implementation. Understanding this interval matters for the evaluation of climate policy: estimates beginning with the first tax payment can miss part of the adjustment that the policy has already induced.

This paper examines how an announcement of future environmental taxation affects current market outcomes. The theoretical starting point is a competitive industry in which changing production plans is costly. A credible future emissions charge reduces the profitability of output at

implementation. Because present and future production decisions are linked, part of that contraction occurs before the charge is collected. The resulting price response depends on demand, technology, adjustment costs, and the credibility and timing of implementation. A model in which producers instead allocate a fixed stock between current and future slaughter gives the opposite prediction. These alternatives make prices informative about possible mechanisms, while also motivating a direct examination of production.

Denmark’s Green Tripartite agreement provides an opportunity to study these predictions. On 24 June 2024, the government and negotiating parties agreed to introduce a tax on livestock greenhouse-gas emissions from 2030 (Danish Government, 2024). The announcement concerned future emissions costs at the farm level. Beef is the focus of the analysis because cattle production involves substantial emissions and biological adjustment lags (Rosen et al., 1994). I compare beef prices before and after the announcement with food products outside the livestock-tax base, and use other countries’ beef markets to examine the sensitivity of that comparison.

I draw on two complementary price sources. Grocery records provide a large product–store panel with detailed descriptions and package information. Statistics Denmark supplies consistently classified monthly consumer price indices. The two difference-in-differences (DiD) estimates indicate price increases of approximately 3 and 6 log points. Synthetic DiD applied to the official indices also yields a positive estimate, with an interval that includes zero. Agreement across the sources is informative about the observed price pattern; the common national policy event means that both remain exposed to the same omitted shocks.

To examine the production margin, I use official monthly bovine slaughter statistics. Denmark is compared with 26 other EU countries, all measured in the same commodity and units. The main synthetic-DiD estimate for slaughter weight is -0.066 log units, with a wide interval. The estimate is close to zero with a longer, seasonally adjusted pre-period. Counts of slaughtered cattle provide similarly imprecise evidence. These results leave a substantial range of production responses compatible with the data and qualify the interpretation of the positive price estimates.

Identification also requires attention to other changes in the Danish cattle sector. In April 2024, the government announced that the cattle derogation from the EU Nitrates Directive would end on 31 July 2024, increasing costs for some cattle farms (Danish Ministry of Environment, 2024). This change overlaps the post-announcement period and can affect the same outcomes as the emissions-tax news. Cross-country comparisons help account for common beef-market developments, but cannot separate these two Danish policy changes. I therefore interpret the estimates as evidence around the announcement episode, conditional on the stated comparison assumptions.

The price evidence raises a further question. If an announcement creates a persistent price signal, how large is that signal relative to the environmental cost that the future tax is intended to address? After presenting the estimates, I examine this question through a literature-based social-cost-of-carbon (SCC) calibration. The exercise retains differences across studies and makes the assumptions about emissions intensity, price bases, persistence, and additional implementation pass-through explicit. Its purpose is to assess magnitudes under stated scenarios; identifying an

optimal tax would additionally require information on emissions, incidence, and welfare.

The paper relates to research on anticipation of environmental regulation (Smulders et al., 2012; Rittenhouse and Zaragoza-Watkins, 2018) and the transmission of emissions costs into product prices (Fabra and Reguant, 2014). Its contribution is to connect an explicit anticipation mechanism to evidence from consumer prices and production before an agricultural emissions tax takes effect. Section 2 develops the model. Sections 3 and 4 describe the data and identifying comparisons. Section 5 presents the evidence, and Section 6 examines its carbon-price equivalent. Sections 7 and 8 discuss what these findings imply for environmental policy.

2 Theoretical Framework

2.1 A credible future tax and planned supply

There are two decision dates: the announcement date, normalized to 0, and the implementation date $T > 0$. Let $Q_t > 0$ denote aggregate beef output at date $t \in \{0, T\}$, let $P_t(Q_t)$ be inverse demand with $P'_t(Q_t) < 0$, and let $C_t(Q_t)$ be the industry's private production cost with $C'_t(Q_t) > 0$ and $C''_t(Q_t) \geq 0$ for $Q_t > 0$. For an arbitrary reference quantity $\bar{Q}_t > 0$, define the competitive-market potential net of production cost as

$$G_t(Q) \equiv \int_{\bar{Q}_t}^Q P_t(x) dx - C_t(Q). \quad (1)$$

The reference quantity changes only an additive constant. Thus $G'_t(Q) = P_t(Q) - C'_t(Q)$ and $G''_t(Q) = P'_t(Q) - C''_t(Q) < 0$. One unit of output generates $\chi > 0$ tons of CO2-equivalent emissions. The statutory tax at T is τ_T DKK per ton of CO2e. Under risk neutrality, the announcement creates the expected output wedge

$$\omega \equiv \chi E_0[\tau_T], \quad E_0[\tau_T] = \pi \tau \quad (2)$$

in the special case in which the tax equals $\tau > 0$ with implementation probability $\pi \in [0, 1]$ and zero otherwise. Here $E_0[\cdot]$ is the expectation conditional on information at announcement. The implementation-date output plan is committed before implementation uncertainty is resolved and cannot be conditioned on the realization of τ_T . Expected tax payments are therefore $\chi E_0[\tau_T] Q_T$.

Production planning is intertemporal. Changing planned output between dates costs $\kappa(Q_T - Q_0)^2/2$, where $\kappa \geq 0$ summarizes biological lags, capacity adjustment, and procurement commitments. Let $\beta \in (0, 1]$ be the one-period discount factor and $\delta \equiv \beta^T$ the discount factor from announcement to implementation. Assume an interior maximizer exists. Because the following potential is strictly concave, that maximizer is unique:

$$G_0(Q_0) + \delta \left[G_T(Q_T) - \omega Q_T - \frac{\kappa}{2} (Q_T - Q_0)^2 \right]. \quad (3)$$

The potential is a device for representing competitive equilibrium. Firms take prices as given, and the first-order conditions equate those prices to private marginal production and adjustment

costs. This interpretation assumes that the aggregate cost and adjustment schedules represent the industry's feasible allocation across firms. Here Q_t denotes planned market output, and κ summarizes the connection between plans across dates. The model abstracts from detailed herd accounting and foreign supply. Applying its price prediction to Danish retail markets consequently also requires that trade and distribution margins do not fully offset the domestic response.

Proposition 1 (Anticipatory contraction) *Suppose the solution to (3) is interior and $\kappa > 0$. A larger expected implementation wedge reduces planned output both before and at implementation and raises the announcement-period market price:*

$$\frac{\partial Q_0}{\partial \omega} < 0, \quad \frac{\partial Q_T}{\partial \omega} < 0, \quad \frac{\partial P_0}{\partial \omega} > 0. \quad (4)$$

The policy-induced level change is zero when the policy is not credible ($\pi = 0$). The marginal pre-implementation response is zero when adjustment is costless ($\kappa = 0$). With fixed adjustment curvature and demand and cost curvatures bounded away from zero, it converges to zero as implementation becomes arbitrarily distant ($T \rightarrow \infty$) when $\beta < 1$.

The proof is in Appendix A. To derive demand and supply from primitives, suppose preferences and technology are stationary across the two dates. A representative household consumes beef $q_t > 0$ and a numeraire good z_t . Its quasi-linear utility is

$$u(z_t, q_t) = z_t + \begin{cases} A q_t^{1-1/\varepsilon} / (1 - 1/\varepsilon), & \varepsilon \neq 1, \\ A \log q_t, & \varepsilon = 1, \end{cases} \quad (5)$$

where $A > 0$ is the taste shifter and $\varepsilon > 0$ is the absolute price elasticity of beef demand. The numeraire price is one and income is sufficient for an interior solution. Utility maximization subject to $z_t + p_t q_t \leq y_t$, where p_t is the beef price and y_t is income, followed by market clearing $q_t = Q_t$, gives inverse demand

$$P(Q) = A Q^{-1/\varepsilon}. \quad (6)$$

Quasi-linearity isolates the commodity-market channel from income effects; it is not required for Proposition 1. The isoelastic schedule is interpreted locally around $Q^* > 0$, rather than as a literal description of willingness to pay near zero consumption.

On the production side, let a representative competitive producer use fixed capital $K > 0$ and a variable composite input $X_t \geq 0$ to produce

$$Q_t = B K^{1-\alpha} X_t^\alpha, \quad B > 0, \quad \alpha \in (0, 1), \quad (7)$$

where B is total factor productivity and α is the output elasticity of the variable input. If that input costs $v > 0$ per unit, conditional cost minimization gives

$$C(Q) = v \left(\frac{Q}{B K^{1-\alpha}} \right)^{1/\alpha}, \quad C'(Q) = m Q^\eta, \quad (8)$$

where $\eta \equiv (1 - \alpha)/\alpha > 0$ is the inverse supply elasticity and $m \equiv (v/\alpha)(BK^{1-\alpha})^{-1/\alpha} > 0$. Here K is quasi-fixed sectoral productive capacity: entry and major capacity expansion are suppressed over the model horizon. The function $C(Q_t)$ captures within-date variable cost conditional on that capacity, whereas $\kappa(Q_T - Q_0)^2/2$ captures additional dynamic reconfiguration costs not contained in X_t . Fixed capacity makes variable cost strictly convex. The no-tax, no-adjustment static equilibrium therefore has

$$Q^* = \left(\frac{A}{m}\right)^{1/(\eta+1/\varepsilon)}, \quad P^* = A(Q^*)^{-1/\varepsilon} = m(Q^*)^\eta. \quad (9)$$

Corollary 1 (Microfounded local incidence) *Under (5)–(8) and $\kappa > 0$, define $s \equiv \eta + 1/\varepsilon > 0$, the normalized adjustment curvature $\rho \equiv \kappa Q^*/P^* > 0$, and the normalized expected wedge $\widehat{\omega} \equiv \omega/P^*$. At the stationary zero-wedge equilibrium $Q_0 = Q_T = Q^*$, the exact local comparative statics are*

$$\left. \frac{\partial \log Q_0}{\partial \widehat{\omega}} \right|_{\widehat{\omega}=0} = -\frac{\delta \rho}{s[s + \rho(1 + \delta)]}, \quad (10)$$

$$\left. \frac{\partial \log Q_T}{\partial \widehat{\omega}} \right|_{\widehat{\omega}=0} = -\frac{s + \delta \rho}{s[s + \rho(1 + \delta)]}, \quad (11)$$

$$\left. \frac{\partial \log P_0}{\partial \widehat{\omega}} \right|_{\widehat{\omega}=0} = \frac{\delta \rho}{\varepsilon s[s + \rho(1 + \delta)]} > 0. \quad (12)$$

Appendix A derives the corollary. Equation (12) is exact for a marginal wedge around the untaxed stationary equilibrium; finite changes obtained from it are first-order approximations, not global effects. Since $\omega = \pi\tau\chi$, the exact marginal response to the announced rate at $\tau = 0$ equals the coefficient in (12) times $\pi\chi/P^*$. Its local marginal magnitude increases with credibility π , emissions intensity χ , proximity to implementation δ , and adjustment curvature ρ . Technology curvature and demand curvature determine the quantity response through s ; $1/\varepsilon$ additionally maps the output response into the price response. The result does not imply that the observed retail-price change equals the future tax wedge.

2.2 Environmental benchmark and a sign-reversing boundary case

Let environmental damages at date t be $D_t(E_t)$, where emissions are $E_t = \chi Q_t$, $D'_t(E_t) > 0$, and D_t is measured in DKK. Define $MD_t^e \equiv D'_t(E_t)$ as marginal damage per ton of CO2e. The standard Pigouvian benchmark is

$$\tau_t^P = MD_t^e. \quad (13)$$

For the paper's accounting exercise, I set $MD_t^e = SCC_t$ under an explicit one-for-one valuation of a ton of CO2e at the CO2 SCC. This convention is not a mechanical equivalence for methane and nitrous oxide merely because global-warming potentials express them in CO2e.

Proposition 2 (Local welfare effect) *Suppose tax revenue is rebated lump-sum, administration*

is costless, private and social discount factors coincide, emissions intensity is fixed, adjustment costs are real resource costs, and the assumptions of Proposition 1 hold. Starting from no emissions price, a small increase in a credible future tax raises welfare whenever marginal environmental damage is positive. More generally, an increase in ω raises welfare locally if the expected output wedge does not exceed marginal implementation-date damage per unit of output, $\omega \leq \chi D'_T(E_T)$; the induced pre-implementation contraction supplies an additional benefit when $D'_0(E_0) > 0$.

The welfare result concerns a marginal policy change from an initially unpriced externality under the model’s assumptions. Assessing the Danish rate requires a further distinction between output and emissions-intensity choices. The announced 2030 schedule has a DKK 300/tCO₂e marginal abatement incentive, while the per-animal deduction produces a DKK 120/tCO₂e average burden at the reference emissions intensity. Both amounts are stated in 2022 prices (Danish Government, 2024; Danish Ministry of Taxation, 2024). For a representative animal with fixed reference intensity, the latter provides the closer approximation to the model’s output wedge. Farms with different emissions per animal face different average burdens.

Finally, the sign in Proposition 1 is not mechanical. Suppose instead that a fixed stock $\mathcal{S} > 0$ of slaughterable output, measured in the same units as Q_t , must be allocated across the two dates, so $Q_0 + Q_T = \mathcal{S}$, with an interior allocation $0 < Q_0 < \mathcal{S}$ and no margin for planned herd contraction. If the tax liability applies to later slaughtered output, a future tax makes later slaughter relatively less attractive, shifts slaughter to date 0, and therefore gives $\partial Q_0 / \partial \omega > 0$ and $\partial P_0 / \partial \omega < 0$. This pure reallocation result need not apply if emissions accrue continuously before slaughter. The preemptive-slaughter channel is the livestock analogue of an anticipation or green-paradox response (Smulders et al., 2012). A positive price response is compatible with planned contraction. The slaughter estimates below examine its quantity implication, while allowing for the distinct timing mechanism in the fixed-stock model.

3 Data and Panel Construction

3.1 Scraped grocery-price microdata

The first source is the Danish grocery-price history assembled by dagligepriser.dk. A raw record identifies a store, product, product description, date, price, and—when available—package quantity and unit. I retain observations with a valid positive price and date. Repeated observations within a product–store–month are averaged, so the unit of observation for the main panel is a product–store pair in a calendar month.

Prices are converted to DKK per kilogram for mass units and DKK per litre for volume units. Reported package quantities take precedence. When quantity or unit fields are absent, the procedure searches the product description for a number followed by kg, g, litre, ml, or cl. Grams, millilitres, and centilitres are converted to their base units before dividing price by quantity. Items measured only per piece and observations with missing, non-positive, or unsupported quantities are

excluded. Because normalized prices remain right-skewed, estimation uses their natural logarithm and Table 1 reports medians and quartiles as well as means.

Product classification is deterministic and auditable. Names and source categories are case-folded; Danish characters and common encoding variants are transliterated; and short tokens are matched at word boundaries. Beef is assigned when the normalized text contains an English or Danish beef term, including *beef*, *okse*, *oksekød*, *kvæg*, *kalv*, *veal*, *entrecôte*, or *roastbeef*. The comparison group consists of classified foods not exposed as livestock treatment items: fish and seafood; fruit and vegetables; grains and bread; plant oils; sweets and snacks; beverages; plant proteins; and pantry goods. Pork, poultry, eggs, lamb, sheep, goat, dairy, and products matching more than one livestock category are never controls for beef. Prepared foods containing terms such as pizza, lasagne, ready meal, or sandwich are flagged as ambiguous when a livestock term is present. Non-food keywords are removed before commodity matching, and unmatched products are excluded rather than silently assigned to the control group.

The event month, June 2024, is excluded from both pre and post. The scraped panel begins in October 2023 and the analysis retains post-announcement months through September 2025, for 23 observed months in total. October 2025, the sixteenth full post-announcement month, and all later months are excluded. The source panel’s final relative-time value is 15; because removing June closes the event-time gap, it is displayed as event month 14. The Danish Veterinary and Food Administration records a bovine viral diarrhoea (BVD) outbreak from October 2025 to May 2026 in three cattle holdings and notes that BVD can reduce appetite and milk output (Danish Veterinary and Food Administration, 2026). A veterinary report states that surveillance detected the affected large dairy herd in September 2025 (Dansk Veterinærtidsskrift, 2025). Ending in September does not prove that all anticipatory disease effects are absent, but it excludes the officially dated outbreak interval and avoids mechanically attributing the subsequent cattle-specific supply disturbance to the tax announcement.

3.2 Official aggregate consumer prices

The second source is Statistics Denmark’s monthly consumer price index by COICOP 2018 product class. Beef and veal (011221) is the treated series. Candidate donors are terminal food and non-alcoholic-beverage classes with complete coverage in the estimation window. I exclude pork; poultry; eggs; lamb, sheep, and goat; dried or salted meat; offal; meat preparations; milk, yoghurt, cheese, cream, dairy desserts, other dairy products; and butter and other dairy fats. These categories are either directly livestock-exposed or sufficiently close to treated animal products that they could absorb the same announcement response. The resulting balanced panel contains beef and 50 donors.

The official-data window runs from April 2023 through September 2025: 15 months before July 2024 and 15 months from July 2024 onward. June 2024 is the final pre-treatment reference month for event-study purposes. This symmetric window prevents the aggregate estimates from being driven by unequal pre- and post-support and, like the microdata restriction, ends before the official

October 2025 BVD outbreak interval.

3.3 Cattle slaughter and production

Eurostat provides monthly bovine slaughter statistics for all 27 EU Member States (Eurostat, 2026). I measure output by carcass weight in thousand tonnes and by the number of animals slaughtered in thousand heads. The main panel covers April 2023–September 2025, matching the official consumer-price window. Each outcome has 810 country–month observations, with Denmark as the exposed country and 26 other Member States as potential comparisons. All countries have positive, non-missing observations throughout this window.

Slaughterhouse activity measures a production margin that can respond to both herd adjustment and the timing of slaughter. It differs from domestic cattle production when live animals cross borders. A separate monthly series from Statistics Denmark measures national cattle production, including the slaughter-equivalent contribution of live-animal trade (Statistics Denmark, 2026). I retain this distinction when interpreting the Eurostat estimates. Longer pre-announcement histories beginning in January 2020 permit a sensitivity analysis accounting for country-specific seasonality.

3.4 EU robustness data

Two robustness checks use the European Commission’s Agri-food Data Portal. The first uses weekly beef-carcass prices reported by EU Member States. I retain the AO2 young-bull series, average prices in euros per 100 kg within country–month, exclude the EU aggregate, and keep the 23 Member States observed in every month from April 2023 through September 2025. Denmark is the exposed country and the other 22 Member States form the donor pool.

The second check uses the European Commission’s Beef Trade Data Explorer, which draws on Eurostat COMEXT data under statistical regime 4. The data report extra-EU beef trade by Member State, external partner, month, and CN8 product. I aggregate imports across beef products to monthly carcass-weight tonnes and balance the panel by coding unreported importer–partner–months as zero. The resulting panel contains 383 importer–partner pairs, including 17 pairs with Denmark as importer, over the same 30-month window.

Table 1 describes the price and slaughter samples. The grocery panel contains 139,332 monthly observations from 15,558 product–store pairs. Product availability is intermittent and normalized price levels are strongly right-skewed, which motivates logarithmic estimation and reporting medians alongside means. The official sample contains 30 monthly observations for each of 51 series. Its tighter distribution reflects the aggregation of price quotes within consistent product classes. Panel C describes monthly slaughter weight and counts in the production estimation sample. Differences in country size make the dispersion across countries informative about scale, while estimation uses logarithms and country-specific comparisons.

Table 1: Descriptive statistics for the estimation samples

Sample	Variable	Observations	Mean	SD	P25	Median	P75
<i>Panel A. Scraped product-store prices</i>							
All	Normalized price	139,332	543.86	4,502.18	50.00	100.00	185.06
Beef	Normalized price	4,555	177.75	644.14	75.00	129.50	214.58
Controls	Normalized price	134,777	556.23	4,575.59	49.95	100.00	183.33
All	Months per product-store	15,558	8.96	6.29	3	7	14
Beef	Months per product-store	535	8.51	5.78	4	7	13
Controls	Months per product-store	15,023	8.97	6.30	3	7	15
<i>Panel B. Statistics Denmark consumer price indices</i>							
All	Consumer price index	1,530	97.82	6.87	95.25	98.76	101.24
Beef	Consumer price index	30	89.43	7.41	84.74	86.58	89.73
Donors	Consumer price index	1,500	97.99	6.75	95.51	98.82	101.26
<i>Panel C. Eurostat monthly bovine slaughter</i>							
All	Weight (1,000 tonnes)	810	19.86	28.10	1.34	5.72	32.78
Denmark	Weight (1,000 tonnes)	30	9.61	0.80	8.91	9.70	10.19
Donors	Weight (1,000 tonnes)	780	20.26	28.57	1.28	4.75	35.17
All	Animals (1,000 heads)	810	67.34	91.41	5.80	18.36	134.28
Denmark	Animals (1,000 heads)	30	35.98	2.99	33.56	36.24	38.14
Donors	Animals (1,000 heads)	780	68.55	92.94	5.57	16.33	147.45

Source: dagligepriser.dk (Panel A); Statistics Denmark (Panel B); Eurostat (Panel C).

Notes: Scraped prices are normalized to DKK per kilogram or litre. For the months-per-product-store rows, observations count product-store units. Panel B covers the balanced April 2023–September 2025 window; its observation count is series-months. Panel C uses the same window and counts country-months. Price donors exclude livestock-exposed food categories; slaughter donors are the other EU countries.

4 Empirical Strategy

The microdata DiD is

$$\log p_{it} = \alpha_i + \lambda_t + \beta(Beef_i \times Post_t) + \varepsilon_{it}, \quad (14)$$

where p_{it} is the normalized price of product–store pair i in month t ; α_i and λ_t are product–store and month fixed effects; $Beef_i$ identifies beef products; $Post_t$ identifies full months after the announcement; β is the average treatment effect on the treated (ATT); and ε_{it} is the error term. Standard errors are clustered by product–store. The event study replaces the post indicator with relative-month interactions and omits May 2024, the final observed pre-announcement month.

For the official-data DiD, I subtract the unweighted mean log CPI of the 50 donors from log beef CPI in each month and regress this gap on a post-announcement indicator. This yields a time-series comparison with 30 monthly observations. Newey–West standard errors allow two lags of serial correlation. The official event study interacts beef with month indicators, controls for calendar month and three- and four-digit COICOP categories, and reports heteroskedasticity-robust standard errors. These event-study intervals describe the pattern under that error specification; they do not account for arbitrary serial dependence within commodity series.

Synthetic DiD follows Arkhangelsky et al. (2021). To use common notation across the balanced panels, let y_{it} denote log CPI, log carcass price, log slaughter output, or log-one-plus imports, as appropriate, and let D_i identify exposed units. With $W_{it} = D_i Post_t$, the estimator solves

$$(\hat{\beta}^{SD}, \hat{\alpha}, \hat{\lambda}) \in \arg \min_{\beta, \alpha, \lambda} \sum_i \sum_t \hat{w}_i \hat{v}_t (y_{it} - \alpha_i - \lambda_t - \beta W_{it})^2. \quad (15)$$

Here α_i and λ_t retain their unit- and month-effect interpretation. Nonnegative donor weights $\hat{w}_i$ align pre-announcement outcome paths with the treated mean, allowing an intercept and regularizing the fit. Nonnegative pre-period weights $\hat{v}_t$ balance donors’ pre- and post-announcement outcomes, also allowing an intercept. Donor weights sum to one, as do pre-period weights; treated units and post-periods receive uniform weights within their respective groups.

Let $\mathcal{C}$ and $\mathcal{T}$ denote donor and treated units, and $\mathcal{P}$ and $\mathcal{A}$ the pre- and post-announcement periods. Write $N_1 = |\mathcal{T}|$, $T_1 = |\mathcal{A}|$, and $\bar{y}_{\mathcal{T}t} = N_1^{-1} \sum_{i \in \mathcal{T}} y_{it}$. The equivalent weighted-gap expression is

$$\hat{\beta}^{SD} = \frac{1}{T_1} \sum_{t \in \mathcal{A}} \left(\bar{y}_{\mathcal{T}t} - \sum_{i \in \mathcal{C}} \hat{w}_i y_{it} \right) - \sum_{t \in \mathcal{P}} \hat{v}_t \left(\bar{y}_{\mathcal{T}t} - \sum_{i \in \mathcal{C}} \hat{w}_i y_{it} \right). \quad (16)$$

Thus the post-announcement gap is adjusted for its weighted pre-announcement level. Uniform donor and pre-period weights recover the balanced-panel DiD comparison. For official CPI, this uniform comparison is the preferred specification in column 2 of Table 2; it uses the consistent official series and accounts for serial correlation in the monthly gap. Synthetic weighting examines sensitivity to the choice of comparison path. The official-panel SDiD uses placebo inference with 200 replications.

Interpreting these coefficients as announcement effects requires that beef and the retained com-

parison foods would have followed parallel trends without the announcement, and that the controls did not respond to it. Excluding livestock products reduces direct exposure to the emissions charge. It does not remove substitution between foods or exposure to the agreement’s other agricultural measures. Product–store clustering also leaves a limitation: a national beef-specific shock can correlate errors across many stores and products. A large number of product–store observations does not provide an equally large number of independent policy assignments. The aggregate specifications offer a complementary view of uncertainty at the commodity level.

I examine two implications using the EU datasets. First, synthetic DiD compares Denmark’s log AO2 beef-carass price with the same series in other Member States. Second, it compares log-one-plus extra-EU imports across importer–partner pairs. These checks use the same 15-month pre- and post-periods. The price specification uses 200 placebo replications. The trade specification uses a 200-replication unit-cluster bootstrap because there are several treated Denmark–partner pairs. All these pairs share a Danish policy exposure, so inference still depends on the assumed dependence structure across pairs. These exercises examine alternative comparisons and an international adjustment margin; they cannot test every spillover or establish parallel counterfactual trends.

The production analysis applies synthetic DiD to log monthly bovine slaughter weight, with Denmark treated and 26 other EU countries as donors. Slaughter counts provide an alternative outcome. Appendix B.1 reports the sample, seasonal adjustment sensitivity, and June-omission check. This design avoids treating another Danish livestock category as unaffected. As in the country-price comparison, a Danish cattle-specific shock remains a potential confound.

In particular, the end of the cattle nitrate derogation was announced on 10 April 2024 and took effect on 31 July 2024 (Danish Ministry of Environment, 2024). It reduced the permitted nitrogen application from livestock manure for affected farms from 230 to 170 kg per hectare. The ministry explicitly anticipated higher costs for some cattle farms. This regulatory change can influence both prices and production during the estimation window. The available data do not identify its contribution separately from that of the livestock-tax announcement.

5 Results

Table 2 reports the price estimates. In the grocery panel, the DiD coefficient is 0.0278, approximately a 2.8 percent increase, with a product–store clustered standard error of 0.0144. Its 95 percent interval includes zero. The official-data DiD is 0.0644, with a Newey–West standard error of 0.0305. Both estimates are positive, although they differ in aggregation, sample composition, and uncertainty. Because they reflect the same national market, treating them as statistically independent would not provide a valid test of equality without an additional covariance assumption.

The official synthetic-DiD estimate is 0.0759, with a 95 percent interval of $[-0.0144, 0.1662]$. Synthetic reweighting thus retains the positive sign but leaves a wide range of magnitudes compatible with the data. Across the three designs, evidence for a positive response is strongest in the

Table 2: Estimated effect of the livestock-tax announcement on beef prices

	(1)	(2)	(3)
ATT	0.0278* (0.0144)	0.0644** (0.0305)	0.0759* (0.0461)
Observations	139,332	1,530	1,530
Units	15,558	51	51
Periods	23	30	30
Pre-treatment average	162.44	84.88	84.88
R^2	0.962	—	—
Time window	2023m10–2025m9	2023m4–2025m9	2023m4–2025m9
Lags only	No	Yes	No

Source: dagligepriser.dk (column 1); Statistics Denmark (columns 2–3).

Notes: The outcome in column 1 is log normalized product price; its DiD includes product–store and month fixed effects. Parentheses contain standard errors: product–store clustered in column 1; Newey–West HAC with two lags for the aggregate DiD in column 2; and the standard error reported by Stata’s synthetic-DiD command under placebo inference with 200 replications in column 3. The observation count in column 2 is the 1,530 series–months in the official panel before cross-sectional averaging into 30 monthly gaps. The pre-treatment average is DKK/kg in column 1 and the CPI index in columns 2–3. Other livestock-exposed products are excluded from every beef control group. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

official time-series DiD. The grocery DiD and aggregate synthetic DiD have p -values of approximately 0.054 and 0.099, respectively, under their reported inference procedures.

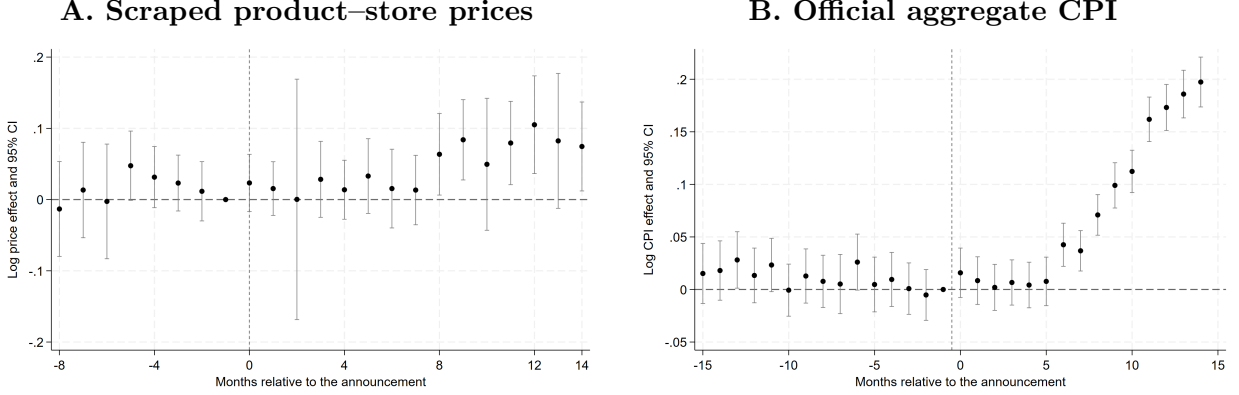
Figure 1 shows how these differences develop over time. The grocery pre-announcement coefficients are jointly insignificant ($p = 0.317$), although the short pre-period limits the power of that diagnostic. Post-announcement prices tend to rise relative to controls with a delay. The official event study also shows a gradual increase. These patterns are compatible with adjustment over time, while an insignificant pretrend test cannot establish the identifying assumption.

The country-price comparison gives an ATT of 0.0448 (placebo standard error 0.2043; $p = 0.826$). Its sign agrees with the consumer-price estimates, but the interval is wide and pre-treatment fit is imperfect (Appendix Figure B3). This comparison adds limited information about magnitude. The extra-EU import estimate is -0.0801 (bootstrap standard error 0.0782; $p = 0.306$; Appendix Figure B4). Both positive and negative import responses remain compatible with its uncertainty. Intra-EU imports are outside this dataset.

The production analysis yields a main slaughter-weight ATT of -0.0661 (placebo standard error 0.0811), or a point estimate of about a 6.4 percent decline. Its 95 percent interval, $[-0.225, 0.093]$, includes zero. With a longer, pre-treatment seasonally adjusted panel, the estimate is -0.0055 . Slaughter counts show similarly uncertain responses. Appendix Table B2 presents these estimates together with the June-omission check. Production evidence therefore does not establish the quantity contraction implied by the model, and the sensitivity across windows is relevant to interpreting the price results.

The negative sign of the main slaughter estimate is nevertheless consistent with the model’s predicted contraction. Its economic magnitude depends in part on supply elasticity: a less elastic

Figure 1: Beef event-study estimates across data sources



Source: dagligepriser.dk (Panel A); Statistics Denmark (Panel B); Eurostat (Panel C).

Notes: Points are relative-month coefficients and bars are 95 percent confidence intervals. In both panels, July 2024 is $t = 0$, the first complete post-announcement month. Panel A omits May 2024 ($t = -1$), because the partial event month of June is excluded, and uses product-store clustered standard errors. Panel B uses June 2024 as its omitted reference month and heteroskedasticity-robust standard errors. Both samples end in September 2025, before the official October 2025 BVD outbreak interval.

production technology limits quantity adjustment to a given change in expected net returns, while more elastic supply permits a larger response. In the model, the inverse supply elasticity η enters together with demand elasticity and intertemporal adjustment costs. A small quantity response alongside rising prices could therefore be consistent with biological or capacity constraints, but the imprecise slaughter estimates do not establish that explanation.

The observed price-quantity combination does not by itself identify supply elasticity. The announcement shifts expected production costs, so the resulting equilibrium moves along demand as well as reflecting supply adjustment. Moreover, consumer prices and slaughterhouse output measure different stages of an internationally traded market. Taking the ratio of the slaughter ATT to the retail-price ATT would consequently not recover a supply elasticity. The evidence leaves both relatively rigid short-run supply and more responsive production compatible with the data; distinguishing them would require farm-level output and producer-price variation that isolates movements along supply.

6 SCC Synthesis and Policy Calibration

The positive price estimates raise a further question: if part of the announcement-period increase persists until 2030, how much of the difference between an environmental-damage benchmark and the future policy's price signal could it cover? Answering this question requires assumptions about persistence, additional price adjustment at implementation, and the emissions base. I first construct a literature benchmark for climate damages, then express it in beef-price units, and finally separate the announcement and implementation components. These calculations describe conditional magnitudes; the reduced-form estimates do not identify the structural parameters needed for

a welfare evaluation.

The benchmark gives equal weight to one selected estimate from each of ten studies (Cai et al., 2016; Nordhaus, 2017; Ricke et al., 2018; Cai and Lontzek, 2019; Pindyck, 2019; Hänsel et al., 2020; van den Bremer and van der Ploeg, 2021; Dietz et al., 2021; Rennert et al., 2022; Barrage and Nordhaus, 2024). This is a documented selection of the literature rather than an exhaustive systematic review. The studies differ in discounting, damage functions, risk, and the object valued. In particular, Pindyck (2019) estimates an average cost of avoiding catastrophic outcomes, which differs from the marginal SCC in the other studies. Excluding that estimate raises the pooled benchmark to USD 166.32/tCO₂. Appendix Table B1 records the chosen estimates and the external comparisons.

Cai et al. (2016) reports both its initial SCC and monetary values for 2010. The inventory treats Pindyck’s monetary base as 2017, an explicit valuation-date assumption; its exclusion sensitivity addresses this additional comparability issue. I inflate dollar values to 2024 US dollars and multiply values per ton of carbon by 12/44 to obtain values per ton of CO₂. The Cai–Lontzek input is the combined economic-and-climate-risk value of USD 124/tC for 2005 in their Table 9, with intertemporal elasticity 1.5 and risk aversion 10. The Barrage–Nordhaus 2030 value is log-interpolated between published 2025 and 2050 points. Only this value and Nordhaus’s reported 2030 estimate refer directly to 2030; their two-study mean is USD 93.36/tCO₂. The remaining values are held constant for the calibration. This is an imposed sensitivity assumption even when a study supplies a rising or stochastic future path. The ten-study mean should therefore be interpreted as a constant-value benchmark used in a 2030 exercise, rather than a pooled forecast for that year.

The equal-paper-weight point mean across ten studies is USD 160.39/tCO₂. The interval is constructed in two explicit layers. Index papers by j . Let θ_j be paper j ’s preferred harmonized SCC estimate and $[L_j, U_j]$ its reported positive central range. Let $1 - 2\alpha_j$ denote its reported coverage where available, or an assumed coverage for a parameter span. Let Φ denote the standard normal cumulative distribution function and define

$$z_j = \Phi^{-1}(1 - \alpha_j), \quad \sigma_j = \frac{\log U_j - \log L_j}{2z_j}.$$

Here z_j is the corresponding standard-normal quantile and σ_j is the calibrated log-scale standard deviation. For a selected paper with a usable range, the within-paper draw is

$$X_j = \exp\left(\log \theta_j - \frac{1}{2}\sigma_j^2 + \sigma_j Z_j\right), \quad Z_j \sim N(0, 1),$$

where Z_j is an independent standard-normal variate, X_j is the simulated SCC value, and $E[\cdot]$ denotes expectation. The $-\sigma_j^2/2$ adjustment ensures that $E[X_j] = \theta_j$, while the range’s log width determines dispersion. Matching the arithmetic mean in this way generally means that L_j and U_j are not the simulated distribution’s corresponding quantiles; the procedure matches their log width, not both endpoints. For papers without a positive range, $X_j = \theta_j$. The 66 percent model range in

Ricke et al. (2018) has $\alpha_j = 0.17$ and therefore $z_j = \Phi^{-1}(0.83)$; the 5th–95th quantiles in Rennert et al. (2022) have $\alpha_j = 0.05$ and $z_j = \Phi^{-1}(0.95)$; and the ethical-parameter range in Hänsel et al. (2020) is assigned 95 percent coverage as an explicit sensitivity assumption, so $\alpha_j = 0.025$ and $z_j = \Phi^{-1}(0.975)$.

In the second layer, each of 10,000 bootstrap replications samples ten papers with replacement from the ten-paper pool, makes an independent within-paper draw for every sampled paper, and records the arithmetic mean of those ten draws. Resampling papers describes sensitivity to the selected pool; the lognormal step adds dispersion calibrated to source ranges. Shared models and assumptions across studies mean that paper draws should not be interpreted as independent scientific replications. The 2.5th and 97.5th percentiles of the 10,000 replicated means are USD 79.50 and USD 366.63/tCO₂. Because the inputs mix model, parameter, and distributional uncertainty rather than comparable sampling errors, this interval is a hierarchical sensitivity envelope, not a frequentist confidence interval for a common structural parameter.

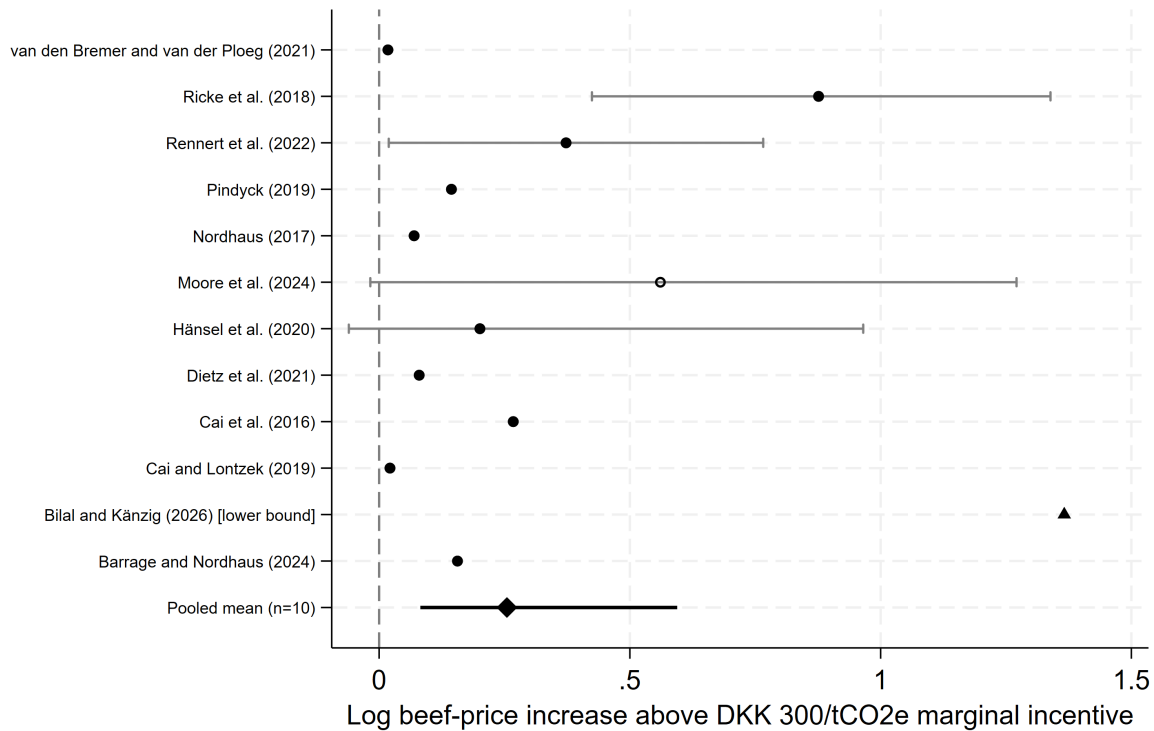
Figure 2 shows each preferred estimate after mapping it into the additional log beef-price increase required above the DKK 300/tCO₂e marginal abatement incentive. Filled points enter the pool. The hollow point is the external literature synthesis, whose inclusion could double count underlying studies (Moore et al., 2024). The partial mortality SCC is reported separately in Appendix Table B1 (Carleton et al., 2022). The triangle is the lower bound reported by Bilal and Känzig (2026) and also remains external. The wide intervals reveal why retaining study-level variation materially changes the calibration.

Let $\bar{p} = 162.44$ be the mean pre-announcement beef price in DKK/kg, $r^{FX} = 6.8953$ DKK/USD the exchange rate, and $\tilde{\chi} = 59.6$ kg CO₂e/kg beef the lifecycle carbon-intensity benchmark. The benchmark follows the OECD’s presentation of the beef-herd estimate from Poore and Nemecek (2018) (OECD, 2025). It includes lifecycle emissions beyond the farm-level emissions base to which the Danish levy applies. Equation (17) is therefore a deliberately broad accounting sensitivity and can overstate the price-equivalent wedge if $\tilde{\chi}$ is interpreted as taxable emissions intensity. For SCC value c^{SCC} in USD/tCO₂ and Danish rate τ in DKK/tCO₂, define the additional log price required as

$$g(c^{SCC}; \tau) = \log \left[\frac{\bar{p} + (c^{SCC} r^{FX} - \tau) \tilde{\chi} / 1000}{\bar{p}} \right]. \quad (17)$$

Here c^{SCC} denotes the SCC input. The quoted statutory rates are DKK 120 and DKK 300 in 2022 prices. Annual Danish net-price indices imply a 2024/2022 factor of 1.04938; hence τ is evaluated at 125.93 and 314.81 in 2024 DKK. The former is the effective average output burden created by the 60 percent per-animal deduction; the latter is the 2030 marginal abatement incentive (Danish Government, 2024; Danish Ministry of Taxation, 2024). At the central calibration, $c^{SCC} = \bar{c}^{SCC} \equiv J^{-1} \sum_{j=1}^J E[X_j]$, where $J = 10$ is the number of eligible papers. Because the within-paper draws are centered so that $E[X_j] = \theta_j$, $\bar{c}^{SCC}$ is the equal-paper-weight mean. In each bootstrap replication, c^{SCC} is replaced by that replication’s arithmetic mean, so the mapped interval uses the same simulated uncertainty as the SCC interval.

Figure 2: SCC estimates and implied beef-price gaps

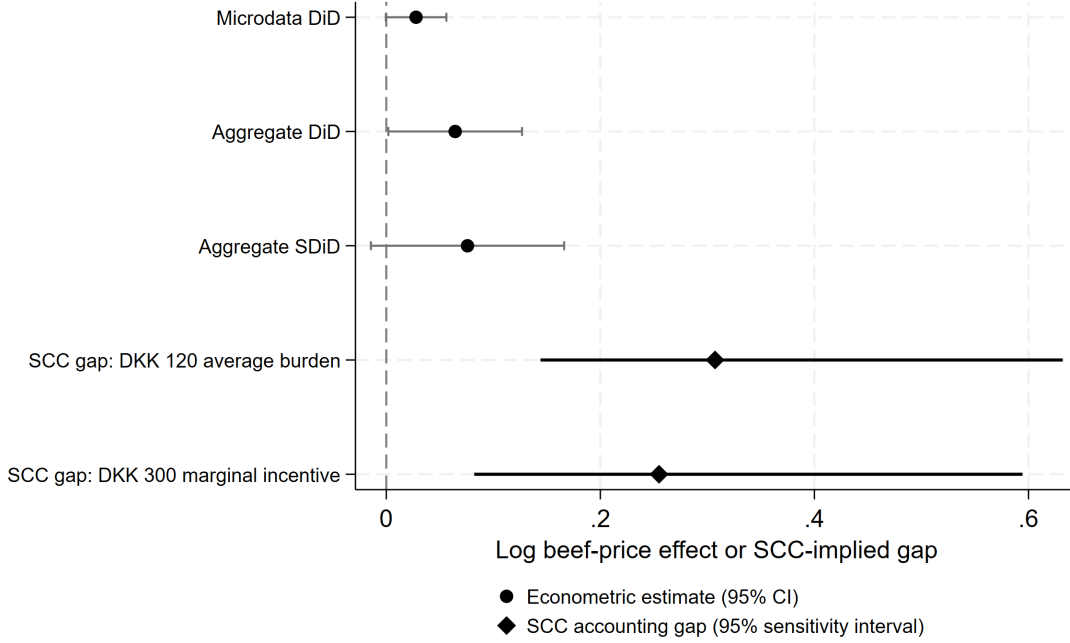


Source: The studies listed in Appendix Table B1; OECD (2025).

Notes: Each point maps a preferred SCC estimate into the log beef-price increase needed above the DKK 300/tCO₂e marginal incentive, using a 59.6 kg CO₂e/kg lifecycle benchmark, 6.8953 DKK/USD, and the scraped pre-announcement beef price. The quoted 2022-DKK statutory rate is converted to 2024 DKK using the annual net-price index. The lifecycle benchmark is broader than the farm-level emissions base subject to the Danish levy, so the mapping is an accounting sensitivity rather than a tax-base calculation. Source ranges preserve their original meaning. Filled circles enter the equal-paper-weight pool; hollow circles and the lower-bound triangle do not. The pooled diamond and line show the hierarchical mean and its 95 percent interval.

At the pooled mean, the log-price gap is 0.307 under the average-burden benchmark, with a sensitivity interval of 0.144–0.632. Under the marginal-incentive benchmark, it is 0.255, with an interval of 0.082–0.594. Figure 3 places these accounting magnitudes alongside the econometric estimates. Their different uncertainty concepts preclude treating interval overlap as a formal test. Across leave-one-study-out calculations, the SCC mean ranges from USD 111.56 to USD 172.36/tCO₂, showing the influence of the highest estimate on the central benchmark.

Figure 3: Econometric estimates and SCC-implied gaps under two policy margins



Source: Statistics Denmark; dagligepriser.dk; the studies listed in Appendix Table B1.

Notes: Points show the three ATT estimates in Table 2 and the SCC-implied log gaps under the DKK 120/tCO₂e effective average output burden and DKK 300/tCO₂e marginal abatement incentive. Quoted statutory rates are in 2022 DKK and are converted to 2024 DKK before calculation. Horizontal bars are 95 percent econometric intervals or the mapped 2.5th–97.5th percentile hierarchical SCC sensitivity interval. The 59.6 kg CO₂e/kg lifecycle benchmark is broader than the levy tax base. These are accounting comparisons, not estimates of the taxable emissions base, tax incidence, or welfare.

To connect the calculation to the question posed at the start of this section, let $a \in [0, 1]$ be the fraction of the estimated announcement increment that persists until implementation, $f \in [0, 1]$ the fraction of lifecycle emissions used as a taxable-intensity approximation, and $\lambda \in [0, 1]$ additional implementation pass-through relative to the reference-animal average burden. The remaining level gap in DKK/kg is

$$R(a, f, \lambda) = \frac{\bar{c}^{SCC} r^{FX} \tilde{\chi}}{1000} - a\bar{p}(e^{\hat{\beta}} - 1) - \lambda\tau_{2024}^{avg} \frac{f\tilde{\chi}}{1000}, \quad (18)$$

where $\hat{\beta} = 0.0644$ is the preferred official-price DiD estimate in column 2 of Table 2 and $\tau_{2024}^{avg} = 125.93$. The last term represents an *additional* price increment after the announcement period. If a pass-through estimate instead measures total incidence including earlier adjustment, that total

must replace the two price terms together. This distinction prevents counting the same adjustment twice. The marginal abatement rate is excluded from this output-price decomposition because the fixed deduction separates marginal emissions incentives from the average output burden.

The CPI estimate is a proportional response rather than a DKK/kg change. Applying it to $\bar{p}$ assumes that the same proportional response applies to the grocery beef-price benchmark. This maps the preferred estimate to an announcement increment of DKK 10.80/kg, while the pooled lifecycle-damage benchmark is DKK 65.91/kg. Under full persistence and no additional implementation increase, the remaining gap is DKK 55.11/kg. With full persistence, full additional pass-through, and the broad assumption $f = 1$, it falls to DKK 47.61/kg.

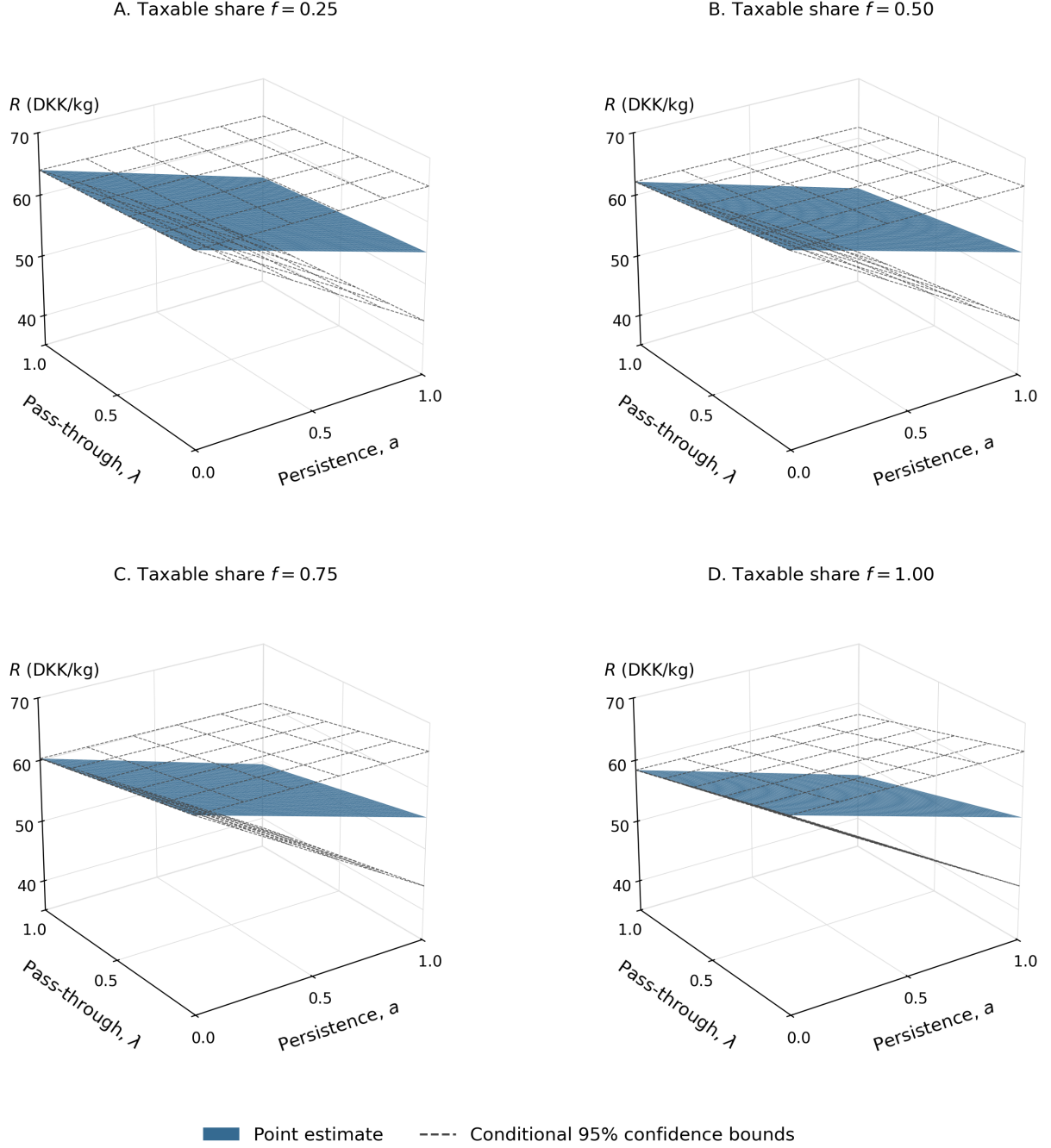
Figure 4 varies announcement persistence a and additional implementation pass-through λ over $[0, 1]$ in four panels, with taxable shares $f \in \{0.25, 0.50, 0.75, 1\}$. Each surface is a plane: persistence subtracts the same announcement increment at every pass-through value, while the slope with respect to pass-through becomes steeper as the taxable share rises. A narrow tax base therefore limits the additional price signal even when pass-through is complete. The panels describe conditional scenarios for implementation, rather than forecasts of these parameters.

The dashed surfaces transform the preferred estimate’s 95 percent HAC confidence interval, $[0.00193, 0.12683]$, through equation (18). Since R decreases in $\hat{\beta}$ for $a > 0$, its lower bound uses the coefficient’s upper endpoint and vice versa. The bounds coincide at $a = 0$, when the announcement estimate no longer enters. This transformation holds the SCC, price benchmark, emissions intensity, exchange rate, and policy rate fixed. It therefore captures sampling uncertainty in the price coefficient conditional on those inputs; the SCC sensitivity interval is a separate source of uncertainty.

Even at $a = f = \lambda = 1$, the conditional 95 percent interval for the remaining gap is DKK 36.44–58.10/kg. The minimum point gap over the entire parameter cube is DKK 47.61/kg, so none of these combinations closes the gap at the central damage benchmark. Full persistence and full additional pass-through together cover about 28 percent of that benchmark. With no additional implementation response, the announcement alone covers at most about 16 percent at the point estimate. Complete closure would require lower environmental damages or a price response beyond the assumed range: at full persistence and pass-through, the SCC threshold is about USD 44.55/tCO₂. A smaller taxable share or weaker persistence lowers that threshold further. These comparisons clarify which assumptions could overturn the conclusion without treating future adjustment as already observed.

The central exercise therefore attributes a modest contribution to the observed announcement-period price movement, conditional on its persistence. Lower SCC benchmarks, a different emissions intensity, or a larger subsequent price response can change that comparison. Moreover, lifecycle CO₂e combines gases with different timing and damage profiles, the pre-announcement grocery mean spans several months rather than a single 2024 valuation date, and Danish production is linked to international markets. The calculation uses a common real-price approximation and a one-for-one CO₂e valuation convention. It cannot establish that the observed price movement

Figure 4: Remaining environmental-price gap at implementation



Source: Statistics Denmark; dagligepriser.dk; the SCC studies in Appendix Table B1.

Notes: The vertical axis is the remaining gap R in DKK/kg from equation (18). The preferred coefficient is the official-price DiD in column 2 of Table 2, applied to the grocery pre-announcement price benchmark. Each panel varies persistence a and additional pass-through λ over their full unit interval; the taxable share f is 0.25, 0.50, 0.75, or 1 as indicated in its title. Solid surfaces give point estimates and dashed meshes transform the coefficient's 95 percent HAC confidence endpoints, holding all other calibration inputs fixed. They exclude uncertainty in the SCC and the assumed mapping from CPI to DKK/kg. All panels share the same vertical scale, which starts above zero; every displayed lower bound is positive.

represents an internalized external cost or identify the optimal Danish tax.

7 Discussion of the Results

The analysis asks whether environmental-policy announcements can influence markets before implementation. The model provides a clear reason for such a response: current production plans adjust when credible news changes expected future costs. The price estimates are consistent with this prediction. Their gradual emergence also fits an adjustment process that operates through production and procurement decisions over time. However, the same pattern can arise from other supply developments, and the available evidence cannot isolate producer expectations.

The production results help assess this interpretation. A contraction in Danish slaughter output relative to other EU countries would support one observable implication of the planned-supply model. The main estimate has that sign, but its uncertainty is substantial and the longer-pre-period estimate is close to zero. The evidence therefore does not confirm the joint prediction of higher prices and lower output. It also does not exclude that prediction. Slaughterhouse activity measures only one stage of production, while herd composition, live-animal trade, emissions intensity, and procurement arrangements may adjust on different horizons.

The international checks serve distinct purposes. Comparing the same beef-carass price across countries reduces reliance on other foods as controls, although imperfect pre-treatment fit limits the information in that estimate. The extra-EU import analysis examines one possible offset to domestic supply changes. Its interval includes economically relevant increases and decreases. Moreover, it excludes intra-EU trade, so it cannot establish the absence of an import response or verify the absence of spillovers into donor countries. Taken together, these exercises broaden the evidence while leaving the core identifying restrictions conditional.

The nitrate-derogation change is a particularly important limitation. It was announced before the livestock-tax agreement and took effect near the beginning of the post-period. Both measures can change expected cattle-production costs. Neither national food comparisons nor cross-country cattle comparisons distinguish their effects with these aggregate data. The broader Green Tripartite package also includes land-use and other agricultural measures. Interpreting the estimates as the isolated causal effect of a future emissions tax would require additional variation in exposure to these measures.

The calibration addresses the size of the observed price movement under explicit assumptions. At the central literature benchmark, the announcement-period price increment covers only part of the gap between environmental damage and the tax-equivalent price signal. This comparison depends on whether the increment persists to 2030, how much further prices adjust at implementation, and which emissions fall within the tax base. The SCC literature also contains substantial disagreement about damage functions, discounting, and uncertainty. These features preclude using the exercise to infer a uniquely appropriate Danish tax rate.

The broader implication concerns credibility as an element of policy design. An announcement

can change investment and production incentives when agents believe that future implementation will occur. The expectations channel has a familiar parallel in Mario Draghi’s *whatever it takes* commitment (Draghi, 2012); in the present setting, it operates through expected production costs and intertemporal adjustment. The model shows that a small credible future charge can improve welfare when emissions are initially unpriced. Establishing that welfare gain empirically would require quantities and emissions as well as prices, and attention to distributional incidence and trade leakage.

Several measurement issues remain relevant. The grocery sample has selective coverage, intermittent product availability, and noisy package normalization. Official CPI reduces some of those concerns but contains a small number of comparison series. Livestock products, including poultry and eggs, are excluded from the food donor pool, yet substitution and exposure to other parts of the agreement can still affect the remaining foods. Finally, surveillance detected the affected dairy herd in September 2025, before the officially dated October BVD outbreak interval (Dansk Veterinærtidsskrift, 2025; Danish Veterinary and Food Administration, 2026). Ending the sample in September reduces exposure to the later outbreak without ensuring that cattle markets were unaffected in the final month.

8 Conclusion

This paper examines adjustment to an environmental tax before its implementation. In a model with costly changes in production plans, a credible future emissions charge reduces planned output and raises prices during the announcement period. Danish grocery prices and official consumer indices show a positive beef-price response following the June 2024 livestock-tax agreement. The estimates from synthetic reweighting are less precise, and the evidence on cattle slaughter does not establish a corresponding contraction in production.

These findings support examining announcements as part of the policy-evaluation window. They also show why the mechanism and the causal attribution require care: production estimates are sensitive to specification, and the end of Denmark’s cattle nitrate derogation overlaps the same period. A literature-based calibration suggests that the observed price increment is small relative to central environmental-damage benchmarks, conditional on persistence and incidence assumptions. Longer panels and farm-level measures of herd adjustment, emissions, and regulatory exposure would help distinguish these channels as implementation approaches.

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A Theory Appendix

A.1 Proof of Proposition 1

The interior first-order conditions for (3) are

$$G'_0(Q_0) + \delta\kappa(Q_T - Q_0) = 0, \quad (\text{A1})$$

$$G'_T(Q_T) - \omega - \kappa(Q_T - Q_0) = 0. \quad (\text{A2})$$

Define $h_0 \equiv -G''_0(Q_0) > 0$, $h_T \equiv -G''_T(Q_T) > 0$, and

$$\Delta \equiv (h_0 + \delta\kappa)(h_T + \kappa) - \delta\kappa^2 = h_0h_T + h_0\kappa + \delta\kappa h_T > 0.$$

Implicit differentiation of (A1)–(A2) gives

$$\begin{bmatrix} h_0 + \delta\kappa & -\delta\kappa \\ -\kappa & h_T + \kappa \end{bmatrix} \begin{bmatrix} \partial Q_0 / \partial \omega \\ \partial Q_T / \partial \omega \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Cramer's rule therefore yields

$$\frac{\partial Q_0}{\partial \omega} = -\frac{\delta\kappa}{\Delta} < 0, \quad \frac{\partial Q_T}{\partial \omega} = -\frac{h_0 + \delta\kappa}{\Delta} < 0. \quad (\text{A3})$$

Since $P'_0(Q_0) < 0$, $\partial P_0 / \partial \omega = P'_0(Q_0) \partial Q_0 / \partial \omega > 0$. The first expression in (A3) is zero when $\kappa = 0$ or $\delta = 0$; when $\pi = 0$, $\omega = \pi\tau\chi = 0$ and there is no policy-induced change. This proves the proposition.

A.2 Derivation of Corollary 1

For $\varepsilon \neq 1$, the marginal utility of beef in (5) is $Aq^{-1/\varepsilon}$; the logarithmic case gives the same expression at $\varepsilon = 1$. Because the numeraire's marginal utility and price both equal one, the household's interior first-order condition is $p = Aq^{-1/\varepsilon}$, which proves (6). Its elasticity is $-(dQ/dP)(P/Q) = \varepsilon$.

With capital fixed, the production constraint (7) requires $X = (Q/(BK^{1-\alpha}))^{1/\alpha}$. Multiplication by the input price v gives (8). Differentiation yields

$$C'(Q) = \frac{v}{\alpha}(BK^{1-\alpha})^{-1/\alpha}Q^{1/\alpha-1} = mQ^\eta, \quad \eta = \frac{1-\alpha}{\alpha} > 0.$$

Equating inverse demand and marginal cost at a static competitive equilibrium gives (9). At that equilibrium,

$$h \equiv -G''(Q^*) = -P'(Q^*) + C''(Q^*) = \frac{P^*}{Q^*} \left(\frac{1}{\varepsilon} + \eta \right) = \frac{P^*}{Q^*} s. \quad (\text{A4})$$

Stationarity and $\omega = 0$ imply $Q_0 = Q_T = Q^*$, so $h_0 = h_T = h$ in (A3). Since $\rho = \kappa Q^* / P^*$, substituting (A4) into (A3), multiplying quantity derivatives by P^* / Q^* , and using $\partial \log P / \partial \log Q = -1/\varepsilon$ gives (10)–(12). These are derivatives evaluated at the nonlinear model's stationary equilibrium,

so no linear demand or marginal-cost schedule is assumed.

Let

$$R(\delta, \rho) \equiv \frac{\delta \rho}{\varepsilon s[s + \rho(1 + \delta)]}$$

denote the coefficient in (12). Direct differentiation gives

$$\frac{\partial R}{\partial \delta} = \frac{\rho(s + \rho)}{\varepsilon s[s + \rho(1 + \delta)]^2} > 0, \quad \frac{\partial R}{\partial \rho} = \frac{\delta}{\varepsilon[s + \rho(1 + \delta)]^2} > 0.$$

Thus proximity and adjustment curvature strengthen local announcement incidence. The factors π , τ , and χ enter through $\hat{\omega} = \pi\tau\chi/P^*$.

A.3 Multi-period anticipation

Consider dates $t = 0, \dots, N$ and implementation at $T \leq N$. Let $\mathbf{1}\{\cdot\}$ denote the indicator function. The multi-period potential is

$$\sum_{t=0}^N \beta^t [G_t(Q_t) - \omega \mathbf{1}\{t \geq T\} Q_t] - \sum_{t=1}^N \beta^t \frac{\kappa}{2} (Q_t - Q_{t-1})^2. \quad (\text{A5})$$

Let $\mathcal{H}$ be its Hessian in $Q = (Q_0, \dots, Q_N)'$ and define $\mathcal{M} \equiv -\mathcal{H}$. The restrictions $G_t'' < 0$, together with the nonnegative quadratic adjustment terms, make $\mathcal{M}$ positive definite. Its diagonal entries are positive, its only off-diagonal entries are non-positive adjacent adjustment terms, and for $\kappa > 0$ it is irreducible. Hence $\mathcal{M}$ is a nonsingular tridiagonal M-matrix and $\mathcal{M}^{-1}$ is elementwise strictly positive. Differentiating the first-order conditions with respect to ω gives

$$\mathcal{M} \frac{\partial Q}{\partial \omega} = -b, \quad b_t = \beta^t \mathbf{1}\{t \geq T\} \geq 0.$$

Therefore $\partial Q / \partial \omega = -\mathcal{M}^{-1}b < 0$ at every date. A credible future wedge thus propagates backward through the production plan.

Under the stationary microfoundations in (5)–(8), evaluate the local response at the zero-wedge equilibrium $Q_t = Q^*$. For interior pre-implementation dates $1 \leq t \leq T - 1$, differentiating the first-order conditions gives the homogeneous recurrence

$$\kappa \psi_{t-1} - [h + \kappa(1 + \beta)] \psi_t + \beta \kappa \psi_{t+1} = 0,$$

where $\psi_t \equiv \partial Q_t / \partial \omega$ and $h = (P^*/Q^*)s$ is given by (A4). Writing the stable component as $\psi_t \propto r^{T-t}$ gives the polynomial $\kappa r^2 - \mathcal{B}r + \beta \kappa = 0$, where $\mathcal{B} = h + \kappa(1 + \beta)$. This polynomial has roots $0 < r_- < 1 < r_+$. Selecting the smaller root is exact for a semi-infinite preperiod under boundedness as $T - t \rightarrow \infty$ and gives an interior approximation for a long finite preperiod:

$$r_- = \frac{\mathcal{B} - \sqrt{\mathcal{B}^2 - 4\beta\kappa^2}}{2\kappa} \in (0, 1), \quad \mathcal{B} \equiv h + \kappa(1 + \beta). \quad (\text{A6})$$

Thus the absolute magnitude of the negative stable component, which is proportional to r_-^{T-t} , declines geometrically with distance from implementation. On an arbitrary finite horizon, the exact path combines both characteristic roots, with coefficients fixed by the initial and terminal boundary conditions; equation (A6) is not an exact global finite-horizon solution. The M-matrix sign result above remains exact. The geometric characterization is the multi-period counterpart to the role of $\delta = \beta^T$ in the two-date model.

A.4 Welfare and the Pigouvian benchmark

Assume tax revenue is rebated lump-sum, tax administration is costless, private and social discount factors coincide, adjustment costs are real resource costs, and emissions intensity χ is fixed so that output is the only abatement margin. With emissions $E_t = \chi Q_t$, social welfare in the two-date model is

$$W = G_0(Q_0) - D_0(\chi Q_0) + \delta \left[G_T(Q_T) - D_T(\chi Q_T) - \frac{\kappa}{2}(Q_T - Q_0)^2 \right]. \quad (\text{A7})$$

At a given date, equating the private output wedge $\chi \tau_t$ to marginal environmental damage per unit of output $\chi D'_t(E_t)$ gives $\tau_t^P = D'_t(E_t) = MD_t^e$, proving (13). The paper's SCC comparison additionally imposes the one-for-one CO2e valuation convention stated in the main text.

Along the private equilibrium defined by (A1)–(A2), differentiation of (A7) and substitution of the first-order conditions yield

$$\frac{dW}{d\omega} = -\chi D'_0(E_0) \frac{\partial Q_0}{\partial \omega} + \delta [\omega - \chi D'_T(E_T)] \frac{\partial Q_T}{\partial \omega}. \quad (\text{A8})$$

Both quantity derivatives are negative by Proposition 1. At $\omega = 0$, $D'_0(E_0) > 0$, $D'_T(E_T) > 0$, $\kappa > 0$, and finite T make both terms strictly positive. More generally, $D'_0(E_0) > 0$ and $\omega \leq \chi D'_T(E_T)$ are sufficient for a positive local welfare effect. This proves Proposition 2. It is a second-best statement about a future instrument: it does not imply that omission of a current tax is first-best.

A.5 Fixed-stock counterexample

Let a fixed stock $\mathcal{S} > 0$ of slaughterable output satisfy $Q_0 + Q_T = \mathcal{S}$, with $0 < Q_0 < \mathcal{S}$. The stock and output share the same units, and the tax liability is assumed to apply to later slaughtered output. Substituting $Q_T = \mathcal{S} - Q_0$, the private objective is

$$G_0(Q_0) + \delta [G_T(\mathcal{S} - Q_0) - \omega(\mathcal{S} - Q_0)].$$

Its first-order condition is $G'_0(Q_0) - \delta G'_T(\mathcal{S} - Q_0) + \delta \omega = 0$. Strict concavity gives

$$\frac{\partial Q_0}{\partial \omega} = -\frac{\delta}{G''_0(Q_0) + \delta G''_T(\mathcal{S} - Q_0)} > 0, \quad \frac{\partial P_0}{\partial \omega} < 0.$$

A future tax therefore accelerates slaughter when fixed stock cannot adjust, opposite to planned contraction. If emissions accrue continuously before slaughter, this two-date result need not apply.

B Additional Results

Table B1: Preferred SCC benchmarks and pooling status

Study	Method	Reference year	2024 USD/tCO ₂	Low	High	Pool status
Cai et al. (2016)	Stochastic IAM	2010	166.9	—	—	Included
Nordhaus (2017)	DICE IAM	2030	74.2	—	—	Included
Ricke et al. (2018)	Spatial IAM	2020	599.9	254.6	1,158.0	Included
Cai–Lontzek (2019)	Stochastic IAM	2005	54.3	—	—	Included
Pindyck (2019)	Expert elicitation	Undated	107.0	—	—	Included
Hänsel et al. (2020)	Updated DICE	2020	133.7	22.5	688.2	Included
van den Bremer–van der Ploeg (2021)	Asset-pricing IAM	2015	52.7	—	—	Included
Dietz et al. (2021)	Meta-analytic IAM	2020	78.5	—	—	Included
Rennert et al. (2022)	Modular IAM	2020	224.2	53.3	500.6	Included
Barrage–Nordhaus (2024)	DICE-2023	2030	112.5	—	—	Included
Carleton et al. (2022)	Mortality damages	2020	21.0	−30.3	65.8	Partial; external
Moore et al. (2024)	Literature synthesis	2020	343.0	38.8	1,059.3	Double-count risk; external
Bilal–Känzig (2026)	Macro empirical	2024	> 1,200	—	—	Lower bound; external

Source: The studies listed in the first column.

Notes: Values are converted to 2024 US dollars per ton of CO₂. One preferred global total-SCC estimate enters per eligible paper. Values at dates other than 2030 are held constant as an explicit sensitivity assumption. Pindyck estimates an average SCC; excluding it gives a mean of USD 166.32/tCO₂. Barrage–Nordhaus is log-interpolated to 2030. Low and high retain their source-specific interpretation. The pooled mean gives equal weight to each included paper; partial estimates, a literature synthesis, and a lower bound are external comparisons.

B.1 Cattle slaughter and production

The model’s production prediction is examined using Eurostat’s monthly slaughterhouse statistics for bovine meat. The primary outcome is the log of carcass weight in thousand tonnes; the second outcome is the log of slaughtered animals in thousand heads. Denmark is the treated unit and the other 26 EU Member States are donors. No other Danish meat category enters the comparison. The statistics measure activity in slaughterhouses (Eurostat, 2026).

The main window matches the official price analysis: April 2023–September 2025, with July 2024 as the first post-announcement month. All 27 countries have positive, non-missing observations in each selected month. Missing values would exclude a country from a balanced specification; they are never replaced by zero. Placebo inference uses 200 replications. The analysis concerns slaughterhouse output, which can differ from domestic cattle production when live animals cross borders or slaughter takes place outside registered facilities. Statistics Denmark’s cattle production series provides a separate national production measure; its live-animal component must be distinguished from the Eurostat slaughterhouse measure (Statistics Denmark, 2026).

Two sensitivity checks examine the dependence on temporal support. The first starts in January 2020, giving 54 pre-treatment months. For each country, it subtracts the pre-treatment mean log output for the corresponding calendar month and adds back that country’s overall pre-treatment mean. This removes estimated country-specific seasonality using only pre-announcement observations. Because it changes both temporal support and seasonal adjustment, it is a joint sensitivity check. The second retains the shorter window but drops June 2024, leaving 14 pre-treatment months. Neither specification is selected on the sign or significance of its estimate.

Table B2: Synthetic DiD estimates for Danish bovine slaughter

	Log slaughter weight			Log slaughter counts		
	(1)	(2)	(3)	(4)	(5)	(6)
ATT	−0.0661 (0.0811)	−0.0055 (0.0803)	−0.0301 (0.0900)	−0.0350 (0.0814)	−0.0225 (0.0788)	0.0020 (0.0924)
Observations	810	1,863	783	810	1,863	783
Countries	27	27	27	27	27	27
Pre-treatment months	15	54	14	15	54	14
Post-treatment months	15	15	15	15	15	15
Long pre-period	No	Yes	No	No	Yes	No
Seasonal adjustment	No	Yes	No	No	Yes	No
June 2024 omitted	No	No	Yes	No	No	Yes

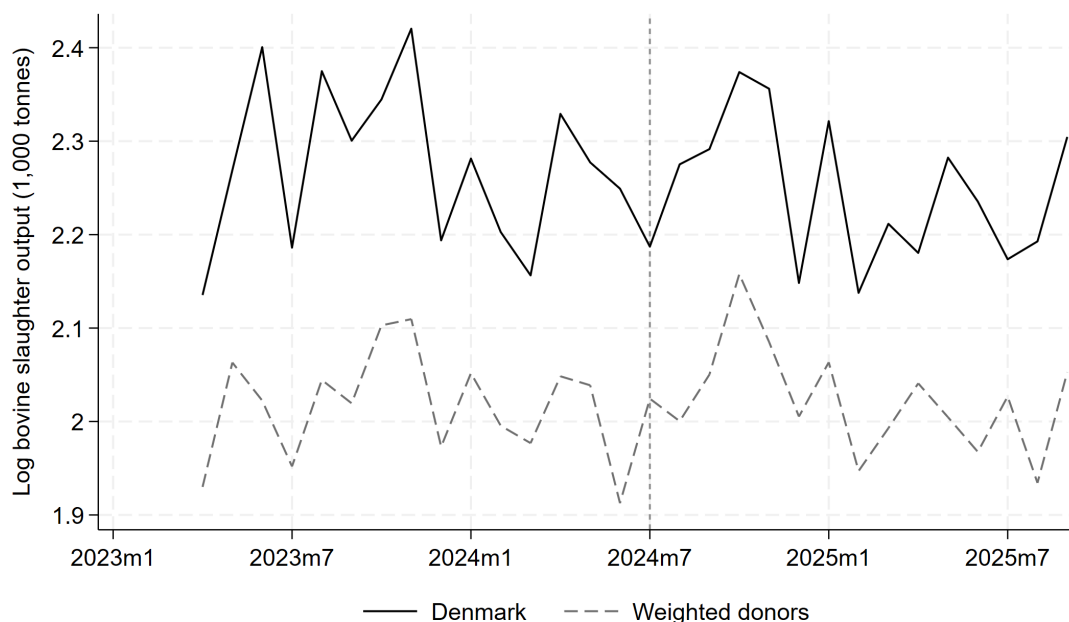
Source: Eurostat, monthly bovine slaughterhouse statistics.

Notes: Denmark is treated and the other 26 EU Member States are donors. Columns 1 and 4 use April 2023–September 2025; columns 2 and 5 begin in January 2020 and remove country-specific seasonality estimated before treatment; columns 3 and 6 omit June 2024. Parentheses contain placebo standard errors based on 200 replications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. No estimate reaches these significance thresholds.

The main weight estimate is negative, but the interval allows a substantial decline as well as an increase. Its pre-treatment centered gap root-mean-square error is 0.056 log units. The longer-

pre-period specification has an error of 0.031 and an ATT close to zero. This sensitivity and the uncertain estimates for slaughter counts limit the evidence for a contraction mechanism. In addition, the end of the Danish cattle nitrate derogation in July 2024 affects the interpretation of every specification. Country controls absorb common market movements but cannot separate that domestic regulatory change from the tax-announcement episode.

Figure B1: Danish bovine slaughter weight and synthetic-DiD comparison

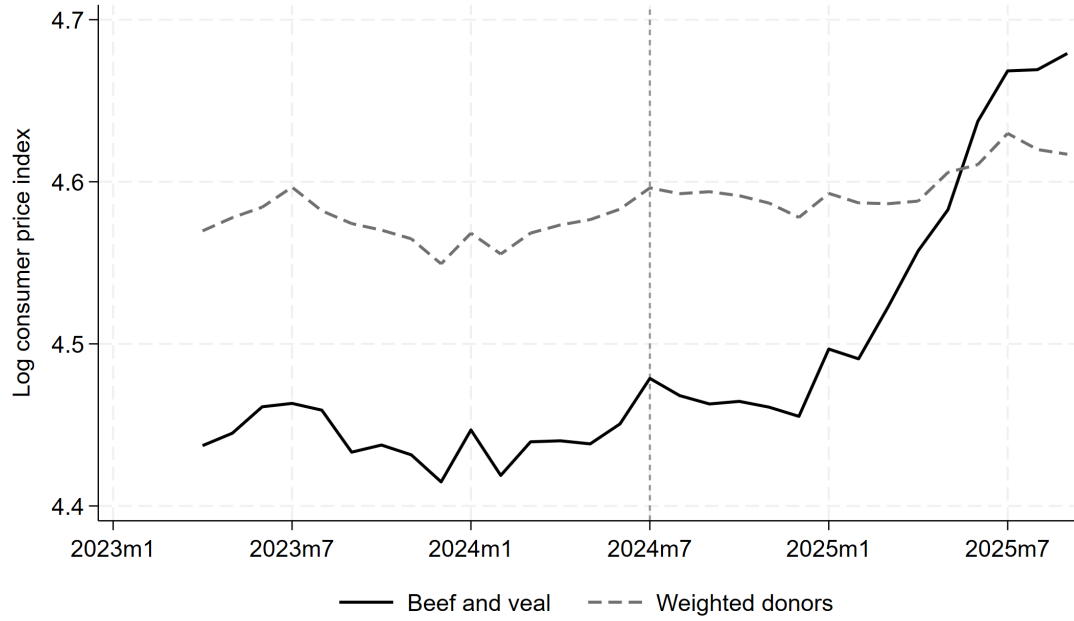


Source: Eurostat, monthly bovine slaughter weight in thousand tonnes.

Notes: The main sample covers April 2023–September 2025. The dashed vertical line marks July 2024. The donor line uses SDiD unit weights and retains the pre-treatment level difference. The ATT additionally uses time weights; the plotted level gap alone is not the estimated treatment effect. Production estimates and uncertainty appear in Table B2.

B.2 Additional price and trade comparisons

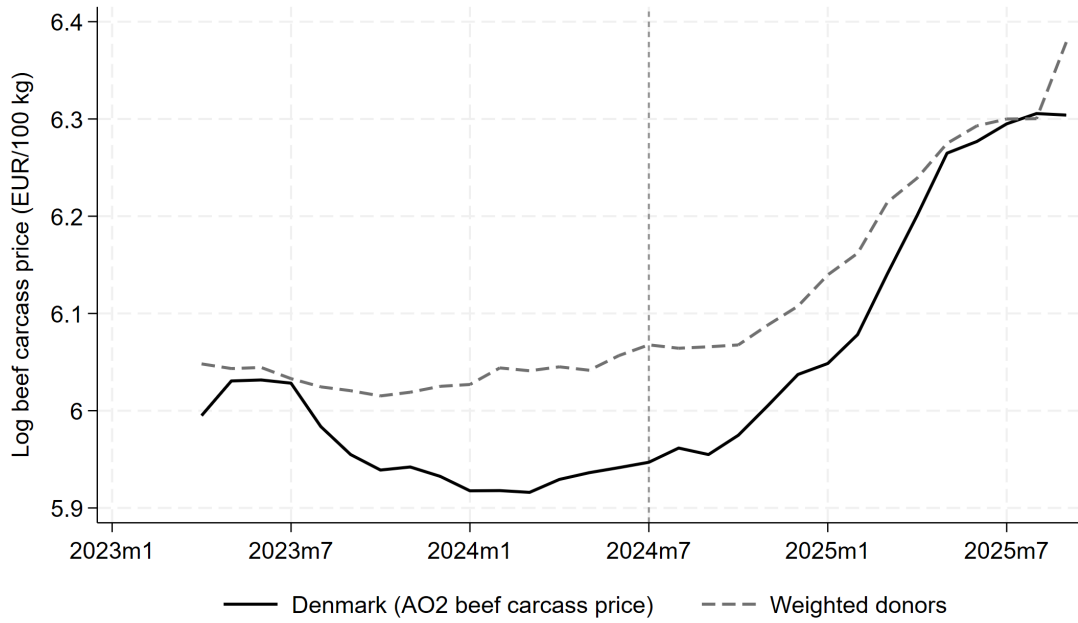
Figure B2: Official-data synthetic DiD: beef and weighted donors



Source: Statistics Denmark.

Notes: Series are log CPI for beef and veal and the SDiD-weighted donor series over the symmetric 15-month pre/post window. The dashed vertical line marks July 2024, the first post-announcement month.

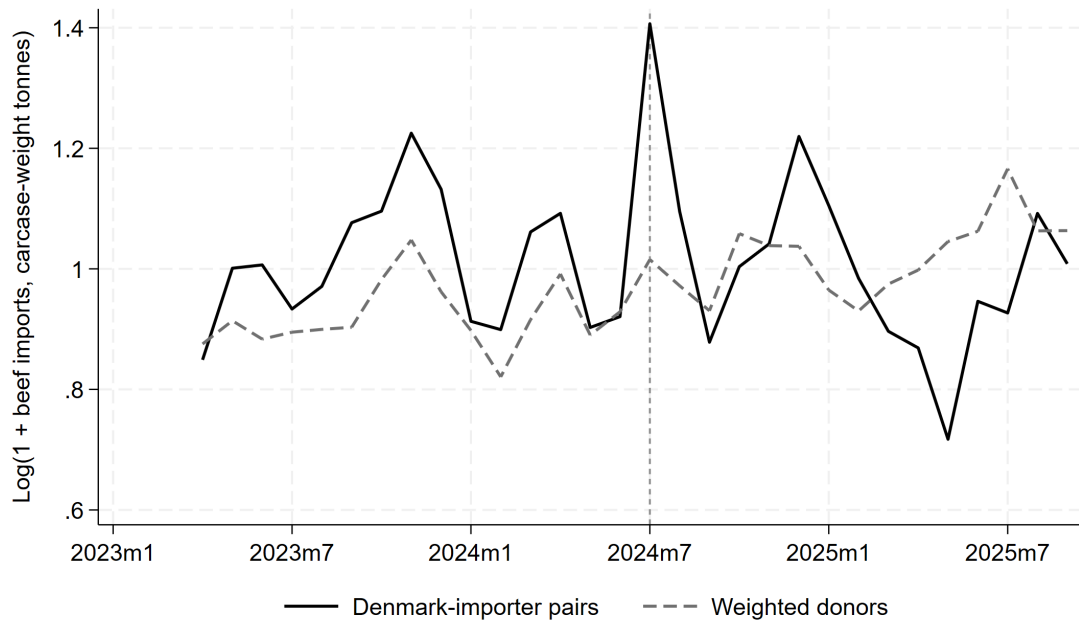
Figure B3: Country-level synthetic DiD: Danish beef-carcass prices



Source: European Commission, Agri-food Data Portal, Beef Prices.

Notes: Series are log AO2 young-bull carcass prices for Denmark and the SDiD-weighted donor series constructed from 22 other EU Member States over the symmetric April 2023–September 2025 window. The dashed vertical line marks July 2024. The pre-treatment fit is imperfect. The ATT is 0.0448 (placebo standard error 0.2043; $p = 0.826$; 200 replications).

Figure B4: Synthetic DiD: beef imports into Denmark



Source: European Commission, Beef Trade Data Explorer; Eurostat COMEXT.

Notes: Series are mean log-one-plus monthly extra-EU beef imports, measured in carcass-weight tonnes, for Denmark-partner pairs and their SDiD-weighted donor series over the symmetric April 2023–September 2025 window. The dashed vertical line marks July 2024. The ATT is -0.0801 (unit-cluster bootstrap standard error 0.0782; $p = 0.306$; 200 replications).